

RITHVIK PRANAO NAGARAJ

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EDUCATION

George Mason University

Master of Science in Computer Science

Kumaraguru College of Technology

Bachelor of Technology in Artificial Intelligence and Data Science

Fairfax, VA

May 2026

Coimbatore, India

May 2024

TECHNICAL SKILLS

Languages: Python, C#, R, JavaScript, SQL, HTML/CSS

Frontend & Backend: React, Node.js, FastAPI, RESTful APIs, Microservices

Machine Learning/Libraries: TensorFlow, PyTorch(CUDA), Scikit-learn, Hugging Face

Databases: PostgreSQL, MySQL, SQLite, DynamoDB

Cloud & DevOps: AWS, Docker, CI/CD

EXPERIENCE

Research Assistant

Jan 2025 – May 2025

George Mason University

Fairfax, VA

- Automated scraping of over 10,000 cybersecurity posts using Python, pandas, BeautifulSoup, and requests, creating a dataset for further analysis
- Built ETL pipelines in Python that cleaned and transformed raw text into JSON datasets, which were used to train large-language models
- Cleaned and organized CVE IDs, exploit titles, and dates with regex and pandas for consistency.

Teaching Assistant (Applied Deep Learning)

Aug 2025 - Dec 2025

George Mason University

Fairfax, VA

- Graded TensorFlow assignments for Applied Deep Learning, providing fast feedback to help students improve.
- Assisted students with debugging models and understanding CNNs, RNNs, and Transformers.

Unity Developer Intern (Blockchain)

Aug 2023 – Nov 2023

Tradala Solution

Chennai, India

- Migrated 2K+ lines of Node.js logic into a Unity SDK in C#, reducing partner integration time by 30%.
- Integrated blockchain transactions with Nethereum to enable smart-contract calls in Unity, making in-game transactions possible and improving user experience.
- Created developer documentation and API references to speed up partner onboarding.

PROJECTS

HealthBridge - Full-Stack Telemedicine Platform

- Designed RESTful APIs using FastAPI for doctor matching, user intake, and ML inference workflows.
- Integrated ML inference (DistilBERT, Whisper) into backend APIs with sub-300ms response latency.
- Built a React + Tailwind CSS frontend consuming backend APIs for real-time user interaction.

MedSAM CUDA-Accelerated Medical Segmentation

- Implemented CUDA-accelerated training pipelines in PyTorch to fine-tune MedSAM on Kvasir-SEG.
- Fine-tuned ViT-based image encoder and mask decoder while freezing prompt encoder per MedSAM design.
- Trained and evaluated multiple models (generalist, fine-tuned, scratch) using Dice+BCE loss.
- Improved Dice score from 0.908 → 0.9448 through domain-specific fine-tuning on GPU.

Secure Artifact Registry — Full-Stack Backend System

- Architected a secure registry for versioned ML artifacts with controlled sharing, reducing data sprawl.
- Built authenticated REST APIs with JWT-based access control enforcing private, shared, and public visibility.
- Designed schemas in SQLite and migrated to PostgreSQL via Docker for production deployment.
- Implemented file upload and metadata pipelines with version tracking, improving reproducibility.

LEADERSHIP & ACHIEVEMENTS

- 2nd Place — Hackathon, George Washington University (2025)